The Fisher–KPP Traveling-Wave Atlas for Every Real Speed: Threshold Fronts, Oscillatory Obstructions, and the Route-A Stop

Route-A structural certificate HCS-C202

Abstract

For all diffusion and growth coefficients $D, r > 0$, we classify every real speed in the Fisher–KPP traveling-wave equation. A dimensionless phase-plane triangle proves existence and translation uniqueness of positive monotone fronts exactly when $c \geq 2\sqrt{Dr}$. Spatial reflection closes the negative-speed family; focus tails exclude all nonzero subcritical speeds, while the stationary Hamiltonian has periodic ovals but no requested heteroclinic. We derive the subcritical, supercritical and repeated-root asymptotics and exclude nonconstant cycles for $c \neq 0$. The Ablowitz–Zeppetella solution supplies an exact control. A finite three-algorithm ledger tests signs and scalings but is never promoted to a continuous shooting proof. Continuous coefficients, heteroclinic rather than periodic carriers, and the absence of a source zeta force all five Route-A gates to fail.

Keywords: Fisher–KPP equation; traveling front; minimal speed; phase plane; heteroclinic orbit; reaction–diffusion.

中文摘要

对任意扩散系数与增长率 $D, r > 0$，本文统一分类 Fisher–KPP 行波方程的全部实速度。 相平面不变三角形证明正单调波前恰在 $c \geq 2\sqrt{Dr}$ 时存在，且除平移外唯一。 负速度由空间反射得到；亚临界速度的焦点尾部必然变号，$c = 0$ 虽有哈密顿周期卵形，却不存在所需异宿轨道。三种临界尾部与精确特殊解共同锁定全部符号和尺度；有限计算只作回归， 不替代连续存在性证明。该模型没有内生素数载体、周期轨道层或目标谱结构，故严格否决路线 A。

关键词：Fisher–KPP方程；行波；最小速度；相图；异宿轨道。

1 Frozen equation and exact scaling

Fix $D, r > 0$ in physical time and consider

$$u_t = Du_{xx} + ru(1-u). \tag{1}$$

Fisher introduced this logistic diffusion model and its wave argument [1]; Kolmogorov, Petrovskii and Piskunov established the classical minimal-speed front theory [2]. With $z = x - ct$, $u(x,t) = U(z)$, (1) becomes

$$DU_{zz} + cU_z + rU(1-U) = 0. \tag{2}$$

Set

$$\xi = \sqrt{r/D}\, z, \qquad s = c/\sqrt{Dr}, \qquad V = U_\xi.$$

The complete parameter family is conjugate, with time orientation retained, to

$$U_\xi = V, \qquad V_\xi = -sV - U(1-U). \tag{3}$$

The equilibria are $(1,0)$ and $(0,0)$. The first is always a saddle; the zero-state characteristic polynomial is

$$\lambda^2 + s\lambda + 1. \tag{4}$$

2 The positive-speed front theorem

Assume $s \geq 2$ and define the slow positive slope

$$q = \frac{s - \sqrt{s^2-4}}{2}, \qquad q^2 - sq + 1 = 0.$$

Consider the compact triangle

$$\mathcal{T}_q = \{(U,V) : 0 \leq U \leq 1,\ -qU \leq V \leq 0\}. \tag{5}$$

On $V = 0$, $V_\xi = -U(1-U) < 0$, while on $U = 1$ one has $U_\xi = V \leq 0$. On the slanted boundary put $G = V + qU$; substitution of $V = -qU$ gives the decisive identity

$$G_\xi = U\{q(s-q) - 1 + U\} = U^2 > 0. \tag{6}$$

The decreasing branch of the one-dimensional unstable manifold of $(1,0)$ enters $\mathcal{T}_q$. Since the divergence of (3) is the negative constant $-s$, a nonconstant periodic omega limit is impossible. The energy identity below also prevents return to the saddle; hence the branch converges to $(0,0)$. It is a profile with

$$0 < U < 1, \quad U_\xi < 0, \quad U(-\infty) = 1, \quad U(+\infty) = 0. \tag{7}$$

Conversely, every orbit with (7) must leave the saddle along this same unstable branch. Reparametrizing one orbit changes only the origin of ξ. Thus for every $c \geq 2\sqrt{Dr}$ the front exists and is unique modulo translation.

3 Every remaining speed

If $W(\xi) = U(-\xi)$, then W solves (3) with speed $-s$. Therefore every $c \le -2\sqrt{Dr}$ has a unique reflected positive profile, increasing from 0 to 1.

When $0 < s < 2$, the roots of (4) are

$$-\frac{s}{2} \pm i\frac{\sqrt{4-s^2}}{2}.$$

Asymptotic linearization at this hyperbolic focus makes every nontrivial tail approaching zero rotate and change the sign of U. It cannot be a $[0,1]$ front. Reflection gives the same obstruction for $-2 < s < 0$.

At $s = 0$, system (3) conserves

$$H_0(U,V) = \frac{V^2}{2} + \frac{U^2}{2} - \frac{U^3}{3}. \tag{8}$$

For every $0 < h < 1/6$, the bounded component of $H_0 = h$ is a periodic oval around the center $(0,0)$; it crosses negative U and is not a positive front. More decisively, $H_0(1,0) = 1/6 \neq 0 = H_0(0,0)$, so no stationary 1-to-0 heteroclinic exists. This closes the full real speed axis.

4 Energy, no cycles, and three leading-edge laws

In physical variables,

$$E = \frac{D}{2}U_z^2 + r\left(\frac{U^2}{2} - \frac{U^3}{3}\right), \qquad E_z = -cU_z^2. \tag{9}$$

Consequently no nonconstant periodic profile exists for $c \neq 0$; this is also Bendixson's conclusion from divergence $-c/D$.

At the back of a positive front,

$$1 - U(z) \sim Ce^{\alpha z}, \qquad C > 0, \qquad \alpha = \frac{\sqrt{c^2+4Dr} - c}{2D} > 0, \quad z \to -\infty.$$

At the leading edge $z \to +\infty$, three distinct laws occur:

$$c > 2\sqrt{Dr}: \qquad U(z) \sim Be^{\lambda_+ z}, \qquad B > 0, \qquad \lambda_+ = \frac{-c + \sqrt{c^2 - 4Dr}}{2D}. \tag{10}$$

$$c = 2\sqrt{Dr}: \qquad U(z) \sim (Az + B)e^{-\sqrt{r/D}\,z}, \qquad A > 0. \tag{11}$$

$$\begin{aligned} 0 < c < 2\sqrt{Dr}: \qquad & U(z) = e^{-cz/(2D)}\{A\cos(\omega z) + B\sin(\omega z) + o(1)\}, \\ & \omega = \frac{\sqrt{4Dr - c^2}}{2D}. \end{aligned} \tag{12}$$

Here $(A,B) \neq (0,0)$ for a nontrivial tail. For (10), the triangle excludes the fast eigendirection. At the critical speed, $W = e^{\sqrt{r/D}z}U$ obeys $W'' = (r/D)e^{\sqrt{r/D}z}U^2 > 0$, giving the positive generalized-eigenvector coefficient A. Equation (12) is the announced sign-changing obstruction, not a numerical shooting observation.

5 Exact control and executable boundary

Ablowitz and Zeppetella found the source-owned member [3]

$$c = 5\sqrt{Dr/6}, \qquad U(z) = \left(1 + e^{\sqrt{r/D}(z-z_0)/\sqrt{6}}\right)^{-2}. \tag{13}$$

Direct differentiation gives zero residual in (2); it tests the speed, length scale and nonlinear coefficient without asserting a closed form at generic speed.

The deterministic release contains 17 speed classes, 340 exact vector-field rows, 25 trapping-boundary rows, six Hamiltonian ovals, nine exact samples of (13), and six physical scalings. The producer-independent Fraction/Decimal checker passes 2,579 assertions; SymPy passes 1,511 checks; replay is byte exact; and 101 repaired-hash attacks (including eight unknown-key injections) plus one stale-hash attack are rejected. Finite rows test implementation only. Classical theory and the written trapping proof carry the continuum quantifiers.

6 Strict Route-A verdict

The coefficients D, r, c and continuous phase coordinates have no intrinsic rational-prime origin. Heteroclinic translation classes are not a primitive periodic-orbit carrier; the profile ODE supplies no source zeta, target divisor, continuation or Weil compression. Its dissipative reduction has no source-native same-clock unitary lift relevant to the target. Hence

$$(A0, A1, A2, A3, A4) = (\mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}),$$

overall `ROUTE_A_REJECTED`, Route B false, under `NO_BAD_EULER_OR_ROOT_NUMBER`.

No target zero/prime table, arithmetic local data, Euler factor, root number, automorphy, target functional equation or Hilbert–Pólya operator is used or claimed. We claim no classical priority, global literature priority, external peer review or acceptance score.

Revision focus. Round 2 adds the exact special-speed control, independent evidence boundary, historical ownership and strict Route-A stop.

Declarations

Data and code availability. Package-local deterministic code and synthetic exact evidence accompany this manuscript.

Ethics. No human, animal, clinical, personal or sensitive data are used; approval is not applicable. Historical sources are cited for mathematical provenance, not ideological endorsement.

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Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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